

Fluoride and Nitrate Geochemistry of Groundwater from Kadiri, Mudigubba and Nallamada Mandals of Anantapur District, Andhra Pradesh, India

B.Muralidhara Reddy¹, V.Sunitha², M.Ramakrishna Reddy³

^{1,2}Department of Geology, Yogi Vemana University, Kadapa, 516 003

³Department of Earth Sciences, Yogi Vemana University, Kadapa, 516 003

¹bandi.murali@yahoo.co.in; ²vangalasunitha@gmail.com; ³reddy.mrk@rediffmail.com

Abstract—Groundwater forms the major source of drinking water in the rural areas of most of the developing nations in the world. Presence of high concentration of fluoride and nitrate in groundwater is a major problem in many countries as it causes health related problems. The present study was carried out with the objective of determining the causes for higher fluoride and nitrate concentration in groundwater in parts of Anantapur District, Andhra Pradesh, India. Fluoride and nitrate concentration in groundwater of this area was analysed by collecting groundwater samples from forty-four representative dug wells. Samples were collected during March 2009–January 2010. The concentration of fluoride in the water samples ranged between 0.1 mg/l and 7.0 mg/l. 76% of the water samples contain fluoride above the maximum permissible limit. Possible sources of fluoride in groundwater are weathering and leaching of fluoride bearing minerals from the basement granitic rocks of this region under alkaline environment. By nature, the granitic rocks of this area possess fluoride values greater than the world average content in granitic rocks. The highest concentration of nitrate recorded during the sampling period was 110 mg/l and the lowest concentration was below detection limit. Taking into consideration 45mg/l of nitrate as the maximum permissible limit for drinking water set by BIS, it was found that 12% of the groundwater samples collected from this study area possessed nitrate concentration beyond the limit. Overall, dug wells present in agricultural fields had nitrate levels within permissible limits when compared to those groundwater samples from dug wells present in settlements which were used for domestic purpose. This indicates that the high nitrate concentration in groundwater of this area is due to leaching from indiscriminate dumping of animal waste. Anthropogenic activities like application of fertilizers for agriculture contribute only little to the groundwater dug well as fluoride. The major contribution of the paper is that the dug wells in the Kadiri, Mudigubba and Nallamada Mandals of Anantapur District, Andhra Pradesh have been demarcated into safe and unsafe dug wells for consumption of water with respect to fluoride.

Keywords—Groundwater; Fluoride; Nitrate; Anthropogenic; Anantapur District; India

I. INTRODUCTION

Groundwater constitutes more than 95% of the freshwater available in the world. Because of the inherent purification by soil, groundwater is generally considered a very good source of drinking water. The quality of ground water is of great importance in determining the suitability of particular ground water for a certain use (public water supply, irrigation, industrial applications etc). The water used for drinking purpose should be free from any toxic elements, living and nonliving organisms and excessive amount of minerals that may be hazardous to health. Fluoride ion in drinking water is known for both beneficial and detrimental effects on health. It is essential for normal mineralization of bones and formation of dental enamel with presence in small quantity [5]. However, when consumed in higher doses ($>1.5\text{mg/l}$), it leads to dental fluorosis or mottled enamel and excessively high concentration ($>3.0\text{mg/l}$) of fluoride may lead to skeletal fluorosis [17]. Contamination of water resources with fluoride beyond acceptable limits is a health associated problem in many areas of South Asia and other regions of the world [9,23]. The amount of fluoride present naturally in groundwater is governed by climate, pH, composition of the host rock and hydrogeology [6]. Some anthropogenic activities also contribute to causing an increase in fluoride concentration in groundwater such as use of phosphatic fertilizers, pesticides, sewage and sludges, depletion of groundwater table etc [18]. Hence to monitor the groundwater quality a lot of studies are carried out throughout the world nowadays [7,11,14,20]. Fluoride is considered as a major pollutant of groundwater on global scale. In the world, around 200 million people from 25 nations have great health risks, with high fluoride in the drinking water [1]. In India, almost 60-65 million people drink fluoride contaminated groundwater and the number affected by fluorosis is estimated at 2.5 to 3 million in many states, especially in Andhra Pradesh, Bihar, Gujarat, Madhya Pradesh, Punjab, Rajasthan, Tamil Nadu and Uttar Pradesh [2,4,13,15,22]. The first case of endemic fluorosis in the country was reported in 1937 in the Prakasam District (erstwhile Guntur District), Andhra Pradesh [19]. The World Health Organization (WHO, 2004) has laid down a limit of 1.5 mg/l fluoride in drinking water. Groundwater is the major drinking water source in the villages of Anantapur district, Andhra Pradesh. The degree of fluorosis has been related to the concentration of fluoride in drinking water and period of consumption. The present paper reports on the prevalence of fluoride concentration in ground waters of revenue Mandals of Kadiri, Mudigubba and Nallamada of Anantapur District, Andhra Pradesh where the people rely on groundwater as potable water source. Critical major inorganic constituents were also determined to ascertain the water quality. Widespread occurrence of nitrate and fluoride contamination in groundwater is reported from different parts of the world. Fertilizer and animal wastes are considered to be the major sources

of this nitrate in groundwater [16]. A high concentration of nitrates (>45 mg/l) may cause methemoglobinemia, gastric cancer and birth defects [12]. To make the water potable these two ions (fluoride & nitrate) have to be treated upon to reduce their concentration. As the groundwater is at a constant threat of being polluted because of natural and anthropogenic sources, the present study was carried out with the objective of understanding the trend of distribution of these ions and their probable sources.

II. METHODOLOGY

A. Study Area

The study area is located in southeastern part of Anantapur District, Andhra Pradesh and lies between longitudes $77^{\circ} 30' - 78^{\circ} 15'$ East and latitudes $14^{\circ} 0' - 14^{\circ} 30'$ North, and falls in the Survey of India toposheets 57 F/14, F/15, J/3, J/4 as shown in Fig.1. The study area mainly exposes peninsular gneisses of Archean age consisting of pink granites, schists, composite gneisses of Dharwar age, intruded by a few younger pegmatite dykes and numerous dolerite dykes and the possible diamondiferous volcanic pipes. The geological map of the study area is shown in the Fig.2. Geomorphic features of the study area include denudational hills, dissected pedimonts, pediplain and flood plain. The climate is semiarid with mean monthly temperatures of January and May are 15°C and 39°C respectively. The normal rainfall of the Anantapur District is 553 mm which is the least rainfall as compared to Rayalaseema and other parts of Andhra Pradesh. The samples were collected from bore wells which were extensively used for drinking and other domestic purposes. The samples were collected in pre-cleaned and sterilized polyethylene bottles of two litres capacity. The depth of the bore wells varied between 250 and 700 feet.

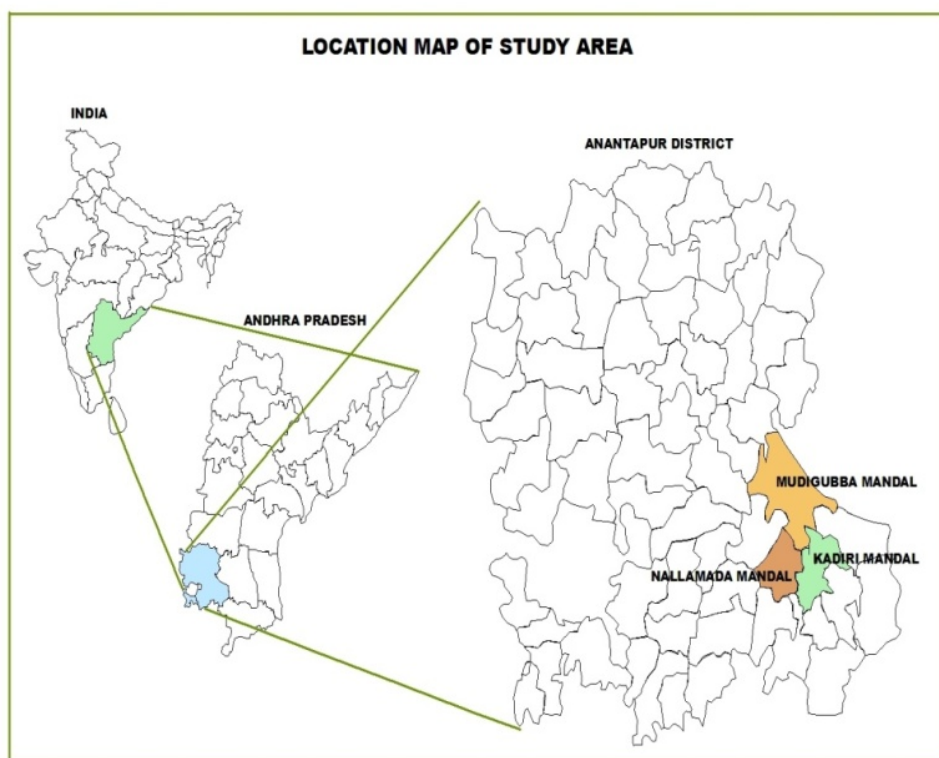


Fig. 1 Location map of Kadiri, Mudigubba and Nallamada Mandals of Anantapur District, Andhra Pradesh, India

B. Hydrogeology

The District is underlain by granite gneisses and schists of the Archaean age. The NE part of the district is covered by consolidated sedimentary formation of Cuddapah Super Group belonging to Late-Precambrian to lower Paleozoic Age. River alluvium occurs essentially along major river courses and to some extent along minor streams. The Archaean crystalline rocks include granites, gneisses and Dharwarian schists. The ground water in these formations occurs in the *weathered and fractured zones* under water table and semi-confined conditions. These rock types do not possess primary porosity. Due to fractures and weathering, they have developed secondary porosity often giving rise to potential aquifers at depth. The degree of weathering in the Archaean formation is less than 20 m. This weathered zone has been tapped extensively by the dug wells and dug-cum-bore wells, which invariably tap the fractures occurring below the weathered zone. Ground water occurring in these formations is generally developed by dug wells, dug-cum-bore wells and bore wells. The depth of open wells ranges from 6.0 to 25.0 m below ground level and depth to water level varies from 1.5 to 23 m bgl. The yield of dug wells varies from 10-200 cm/day for a pumping period of 3 to 6 hrs/day.

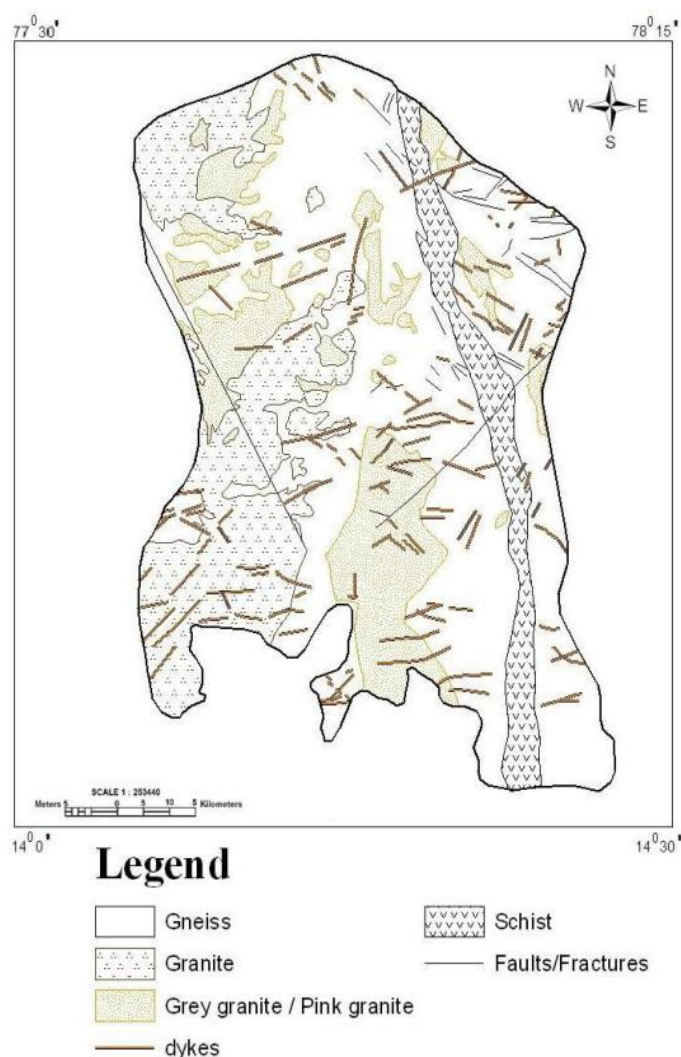


Fig. 2 Geological map of Kadiri, Mudigubba and Nallamada Mandals of Anantapur District, Andhra Pradesh, India

III. RESULTS AND DISCUSSION

The analysis of groundwater samples collected during the period from March 2009 to January 2010, indicates that the concentration of fluoride and nitrate varies with respect to space and time. The detailed description of the causes for this variation is discussed in the following sections. The maximum and minimum levels of groundwater levels and values of pH of groundwater samples are given in Table I.

TABLE I GROUNDWATER LEVELS AND REPRESENTATIVE PH VALUES OF GROUNDWATER SAMPLES OF KADIRI, MUDIGUBBA, NALLAMADA MANDALS OF ANANTAPUR DISTRICT, ANDHRA PRADESH

Month	Groundwater level (bgl in m)		pH	
	Min	Max	Min	Max
March 2009	0	12.0	7.0	8.5
May 2009	0	9.0	7.0	8.8
July 2009	0	11.0	7.2	8.5
September 2009	0	10.8	7.5	9.2
November 2009	0	9.5	6.9	8.4
January 2010	0	11.5	7.2	8.5

A. Fluoride

The fluoride concentration in the study area during the period of study varies from 0.1 mg/l to 7.0 mg/l. The minimum, maximum and mean of fluoride concentration during the sampling months are given in Table II.

TABLE II STATISTICAL SUMMARY OF FLUORIDE (F) CONCENTRATION (Mg/L) IN THE KADIRI, MUDIGUBBA, NALLAMADA MANDALS OF ANANTAPUR DISTRICT, ANDHRA PRADESH

Month	Number of samples	Min (mg/l)	Max (mg/l)	Mean (mg/l)
March 2009	40	0.2	6.9	3.61
May 2009	35	0.5	7.0	3.66
July 2009	44	0.1	6.8	3.60
September 2009	42	0.3	5.5	3.53
November 2009	40	0.1	6.2	2.98
January 2010	44	0.6	7.0	3.66

The World Health Organization [25] has specified the desirable limit in fluoride in drinking water to be between 0.5 to 1.5 mg/l. Accordingly, the groundwater samples of the study area were classified based on their fluoride concentration as shown in Table III. This study shows that out of the total samples analysed, 76% of the samples are above the maximum permissible limit. The pH of groundwater ranges from 6.9 to 9.2. Most of the groundwater in this area is alkaline in nature. It was observed that an increase in the alkalinity value made a similar increase in the amount of fluoride. In alkaline environment, the fluoride ion can be easily liberated from their ore because OH⁻ and F⁻ ions have similar radii. Hence they easily exchange with each other. [9,13,24,26].

TABLE III WHO CLASSIFICATION OF GROUNDWATER OF KADIRI, MUDIGUBBA AND NALLAMADA MANDALS OF ANANTAPUR DISTRICT, ANDHRA PRADESH BASED ON FLUORIDE CONCENTRATION

Month	"Less than the permissible" range (<0.6 mg/l)	"Permissible" range (0.6-1.5 mg/l) (%)	"Greater than the permissible" range (>1.5 mg/l) (%)
March 2009	12	16	72
May 2009	4	18	78
July 2009	6	15	79
September 2009	11	14	75
November 2009	10	16	74
January 2010	11	15	74

Anthropogenic activities such as fertilizers may also lead to increase in fluoride content in groundwater. However, contribution of fluoride to groundwater by these activities is negligible amounts only when compared to the contribution by weathering of rocks. The concentration of fluoride in groundwater samples taken from wells in agricultural fields and domestic wells was compared and it was found that they did not show much difference. This indicates that the use of fertilizers is not a major source for fluoride in groundwater in this area. The average fluoride concentration in the groundwater samples show that 13 dug wells out of 44 are safe for drinking purpose whereas 31 dug wells are unsafe. Hence, steps should be taken to provide alternate source of drinking water for the people who depend on these wells for their daily needs or the quality of groundwater has to be improved by adopting suitable methods.

B. Nitrate

The statistical summary of the results of nitrate concentration present in the groundwater samples analysed is given in Table IV. In this study the nitrate concentration in groundwater ranges from 110 mg/l. 88% of the samples had nitrate concentration above the maximum permissible limit of 45 mg/l as prescribed by BIS (1992). The percentage of groundwater samples having nitrate concentration higher than maximum permissible limit of 45 mg/l during each sample is given in Table IV.

TABLE IV STATISTICAL SUMMARY OF NITRATE CONCENTRATION (Mg/L) IN THE GROUNDWATER OF STUDY AREA DURING VARIOUS MONTHS

Month	Number of samples	Min (mg/l)	Max (mg/l)	Mean (mg/l)	%above 45 mg/l
March 2009	40	20	110	65.94	88
May 2009	35	20	100	61.67	85
July 2009	44	33	240	74.62	87
September 2009	42	30	110	66.00	82
November 2009	40	20	100	61.67	83
January 2010	44	20	110	65.94	85

As agriculture is one of the main activities of this region, a considerable portion of applied agro chemicals are expected to penetrate the soil and reach the groundwater table. This is likely to increase nitrate (NO₃⁻) level in groundwater. But the concentration of nitrate in the groundwater samples collected from wells in agricultural fields of this study area was found to be within the permissible limits of 45 mg/l. Of the total twenty samples wells tested in the *agricultural fields*, five wells showed high nitrate concentration i.e. above 45 mg/l. This shows that even though irrigation is the major practice in this area, there is no significant influence of this activity on the nitrate contents of groundwater. Of 24 wells present in *domestic areas*,

15 of them showed high nitrate levels at least during one of the sampling months. Hence, it is felt that the elevated nitrate concentrations in the samples of the groundwater are due to the intensive use of fertilizers in the study area. Leaching of nitrate to groundwater is mainly due to excessive application of N-fertilizer, the absence of proper soil and water management practices, septic tanks, improper disposal of domestic wastes.

C. Managing Groundwater Quality

Groundwater in the study area is polluted with fluoride and nitrate. Defluorinating and denitrifying the groundwater with the help of cost-effective methods needs to be practiced in large extent to reduce the fluoride and nitrate concentration of groundwater before using it for domestic purposes.

Nitrate (NO_3^-) contamination of the groundwater, due to the intensive use of fertilizers has become a serious ecological problem in the study area. Leaching of nitrate to groundwater is mainly due to excessive application of N-fertilizer, the absence of proper soil and water management practices, septic tanks, improper disposal of domestic wastes. Further, rain water harvesting in these regions can help to a great extent in diluting the fluoride and nitrate concentration in groundwater.

IV. CONCLUSIONS

This study was carried out with the aim of determining the groundwater quality in parts of Anantapur district, Andhra Pradesh with respect to fluoride and nitrate. Of the total 235 samples analysed, 16% of the groundwater samples are within the range of 0.5 to 1.5 mg/l of fluoride and hence they are suitable for drinking. However, 8% of the samples are below the permissible limits and 76% are above 1.5 mg/l of fluoride. The nitrate concentration was high i.e. above the maximum permissible limit of 45 mg/l in 88% of the samples. The present study concludes that alkaline environment is the dominant controlling factor for leaching of fluoride from the source material in the groundwater. Evapotranspiration, longer residence time of water with the aquifer material, the dissolution of fluorite, apatite and topaz from local bed rocks and usage of agricultural fertilizers, the absence of proper soil and water management practices, septic tanks, improper disposal of domestic wastes, animal wastes, sewage dump lead to high fluoride, nitrate concentration in groundwater.

The most effective way of reducing the nitrogen content of groundwater in the areas where agriculture is the main occupation is to reduce the application of fertilizers in consultation with agriculture scientists and change the cropping pattern by going in for irrigated dry crops which consume less water and fertilizers. It is suggested that frequently modifying the cropping sequences offer possible ways to scavenge the nitrogen and provide fluoride-free drinking water in the study area.

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B. Muralidhara Reddy did B.Sc in Loyola Degree college, Pulivendula, Kadapa District, Andhra Pradesh in 2006 and M.Sc Geology from S.V University, Tirupati in 2008. Presently he is working for Ph.D in the Department of Geology, Yogi Vemana University, Kadapa and also working in the DST project. He has published eight International and eight National Research Publications and participated in five national workshops and presented ten papers in national seminars.



Dr. V. Sunitha, did B.Sc in Govt Arts College, Anantapur District. Andhra Pradesh. She did M.Sc Geology from S.V University, Tirupati in 1997 and M.Sc Tech Environmental Geochemistry from Osmania University, Hyderabad in 2000. She Completed her Ph.D in Applied Geochemistry from Osmania University Hyderabad in 2004. She did her post Doctoral Research at Osmania University and **National Geophysical Research Institute**, Hyderabad. She has been awarded with Young Scientist Award in 2008, from **DST** (Govt. of India), New Delhi, India. She published thirty five National and International research papers. Her major research fields are Hydrogeochemistry, Remote Sensing and GIS. At present she is working as Assistant Professor, Department of Geology, Yogi Vemana University, Kadapa, Andhra Pradesh, India.



Prof M. Ramakrishna Reddy did M.Sc Geology from S.V University, Tirupati in 1980. He has published twenty publications in National and International Journals and participated in several National and International seminars and workshops. He received Prestigious A.P Best Teachers Award in 2011. Presently, he is working as Registrar of Yogi Vemana University, Kadapa.